> 2026 MIDWEST BIOINFORMATICS TRAINING

TUESDAY & WEDNESDAY, MAY 12-13, 2026

MDHHS Bureau of Laboratories
Lansing, Michigan

> MIDWEST BIOINFORMATICS TRAINING

FINAL AGENDA

TUESDAY, MAY 12

MDHHS Bureau of Laboratories
3350 N. MLK Jr. Blvd, Lansing, MI
Room 1029-B

Arrival & breakfast — ***08:00 am***

Welcome & introductions — **08:30 am**
Macy Pell, PhD | Arianna Miles-Jay, PhD, MPH | Heather Blankenship, PhD

Bioinformatics in public health — **09:15 am**
Heather Blankenship, PhD | Arianna Miles-Jay, PhD, MPH

Training overview: From raw reads to phylogenetic trees — **09:45 am**
John Chodkowski, PhD

Introduction to GitHub & Codespaces demo — **10:05 am**
John Chodkowski, PhD

Morning break — ***10:15 am***

Training module 1: Getting to know the pipeline — **10:30 am**

Assessing read quality with FastQC — **10:45 am**
John Chodkowski, PhD

Read processing and trimming with fastp — **11:15 am**
John Chodkowski, PhD

Activity: Interpreting sequence quality — **11:30 am**
John Chodkowski, PhD

Lunch break — ***12:00 pm***

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Training module 1: Getting to know the pipeline (continued)

From fragments to function: Genome assembly methods — **01:00 pm**
Douglas Maldonado-Torres, MS

Activity: Hakuna Matata, it means no worries (about assembly) — **01:30 pm**
Ellie Yoon, MS

Assessing assembly quality with QUAST — **02:15 pm**
Douglas Maldonado-Torres, MS

Afternoon break ---------- ***02:30 pm***

Training module 2: End-to-end pipeline & reference-based assembly — **02:45 pm**

Putting it all together: End-to-end pipelines — **02:50 pm**
Douglas Maldonado-Torres, MS

Reference-based genome assembly — **03:00 pm**
Douglas Maldonado-Torres, MS

Activity: Interpreting assembly quality — **03:15 pm**
Douglas Maldonado-Torres, MS

Knowledge check & review — **04:00 pm**
Macy Pell, PhD

Day 1 wrap-up & conclusions — **04:20 pm**
Macy Pell, PhD

Day 1 concludes ---------- ***04:30 pm***

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WEDNESDAY, MAY 13

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Arrival & breakfast — ***08:00 am***

Day 2 overview — **08:30 am**
Macy Pell, PhD

The alphabet soup of molecular typing — **08:40 am**
Arianna Miles-Jay, PhD, MPH

Training module 3: Bacterial whole genome sequencing analysis — **09:10 am**

MLST typing — **09:15 am**
Karla Vasco, DVM, MSc, PhD

Variant calling — **09:30 am**
John Chodkowski PhD

Morning break — ***09:45 am***

Core SNP alignment — **10:00 am**
Douglas Maldonado-Torres, MS

Constructing a phylogenetic tree — **10:20 am**
Karla Vasco, DVM, MSc, PhD

Microreact tool demo — **10:40 am**
Sara McNamara, MPH, MLS(ASCP), CIC

Activity: Visualizing phylogenetic trees using Microreact — **11:00 am**
Karla Vasco, DVM, MSc, PhD | Sara McNamara, MPH, MLS(ASCP), CIC

Lunch break — ***11:30 am***

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Training module 3: Bacterial whole genome sequencing analysis (continued)

Genome annotation — **12:30 pm**
Karla Vasco, DVM, MSc, PhD

Pangenome analysis with Panaroo — **01:00 pm**
Karla Vasco, DVM, MSc, PhD

Activity: Visualizing pangenome analysis outputs and rooted trees — **01:20 pm**
Karla Vasco, DVM, MSc, PhD

Afternoon break ---------- ***01:45 pm***

Tools, tips, & tricks: Q&A panel with bioinformatics experts — **02:00 pm**
Moderator: Macy Pell, PhD
John Chodkowski, PhD - Bioinformatics Specialist
Douglas Maldonado-Torres, MS - Bioinformatician
Karla Vasco, DVM, MSc, PhD - Bioinformatics Specialist
Ellie Yoon, MS - Bioinformatician

Exploring best practices: Introduction to Nextflow — **02:45 pm**
John Chodkowski, PhD

Training evaluation & concluding remarks — **03:30 pm**
Macy Pell, PhD | Arianna Miles-Jay, PhD, MPH | Heather Blankenship, PhD

Day 2 concludes ---------- ***04:00 pm***